

Lab 07

Due March 21 5:00 PM PST

STAT240 Spring 2025

2025-03-17

For this homework, provide a single rendered Quarto Markdown file in pdf format on crowdmark for the problems (you may render the QMD file to html, and then convert the html file to pdf using the print function on your web browser). You can use the template `Lab_07.qmd`. **Provide in the markdown the final version of all of the code you wrote for this homework, and make sure long lines of code are wrapped in the rendered pdf. Lines which are cut off cannot be graded. If your Markdown file involves examining a large dataset, do not print the entire dataset to the markdown file in the steps of your solution (instead, suppress the output, or only show a small section of the data as an example).**

Problem 1 (Clustering)

In class last week we wrote an R function `my_dist`, to compute the distance between points and a set of centroids. This is a key component in the implementation of k-means clustering. In this problem we will build on that code to implement k-means clustering from scratch.

Problem 1a

Solve the first Exercise problem in last weeks slides, where you write an R function `my_dist` which will return the distance between n data points and K centroids in an $n \times K$ matrix.

(2 points)

Problem 1b

Write your own implementation of the k-means without using any libraries and without using the R function `kmeans`, as a function called `my_kmeans`. It should work for datasets of arbitrary dimension D , and should return the centroids and the cluster assignments of the last iteration of the algorithm. You may work from the code from the slides for Week 9 (but extend from 2 dimensions to arbitrary dimension, and use your own `my_dist` function). For stopping condition, have your function take two parameters (in addition to a parameter for the data):

- A maximum number of iterations such that if the number of iterations reaches this maximum number, the iterations stop,
- A threshold such that if the all Euclidean distances between the centroids before an iteration and after an iteration is less than this threshold, the iterations stop.

Set reasonable default values for these parameters. Provide your code, and write a one paragraph help for your function, indicating the names of the parameters, and the nature of the return value.

(5 points)

Problem 1c

Now implement your clustering method for `peng_scaled` with $K = 3$ clusters. Choose the maximum number of iterations to be 100 and the threshold to be 0.001.

Compare the clusters from your implementation to those from the `kmeans` function, with the corresponding values. Do they return the same clustering assignments? How can you tell?

(3 points)